Microsoft Azure Resource Manager: An Evaluation of Cloud Deployment and Infrastructure Design

Module: Cloud Operations and Management

Coursework: Group Report A

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Overview

Azure Resource Manager (ARM) is the deployment and management service for Microsoft Azure (Microsoft, 2026b). It provides a single, centralised control plane through which organisations create, modify, and delete their resources, such as virtual machines, networks, storage, and databases, consistently and, where appropriate, in parallel, regardless of the tool used.

ARM is characterised by several features. It integrates Role-Based Access Control (RBAC), which governs who may manage resources within an Azure subscription, thereby strengthening security and supporting compliance. Its operations are declarative, since the user specifies the desired end state rather than the individual steps, and idempotent, meaning a deployment can be repeated without altering the outcome. Resources are organised into resource groups according to their lifecycle, while tags and policies support governance and the enforcement of rules.

Infrastructure is defined as code (IaC) using ARM templates, JSON files specifying the resources to deploy, which allows infrastructure to be defined, versioned, and deployed reproducibly (Rahman et al., 2019).

Service Model Analysis

Customer control decreases from IaaS to SaaS (Younis et al., 2024), while ARM directly manages Azure IaaS and PaaS resources and has a more limited role for SaaS (Microsoft, 2026b). Under Infrastructure as a Service (IaaS), ARM provisions the underlying infrastructure. The organisation gains maximum control, but also maximum responsibility. ARM automates provisioning, yet the operating system, patching and runtime security remain the customer's burden.

Under Platform as a Service (PaaS), ARM provisions managed services in which the platform is maintained by Azure, allowing developers to focus on code and deployment, for example Azure App Service or Azure SQL Database. The organisation trades configuration control for reduced operational effort, an exchange that suits application teams but constrains workloads needing OS level access.

Under Software as a Service (SaaS), the contrast is greatest. The application is fully managed and consumed directly. Under SaaS, the provider manages the application and underlying infrastructure, so ARM has limited direct control over the SaaS application itself.

RBAC, tags and Azure Policy can support consistent governance across many IaaS and PaaS resources, subject to the capabilities of each resource type.

Deployment Types

Cloud deployment models determine where resources are hosted and controlled. NIST defines four deployment models, including public, private and hybrid models (Mell and Grance, 2011). For the proposed secure web application, a public-cloud deployment would host its compute, storage, database and networking resources in Microsoft Azure. Azure Resource Manager (ARM) acts as Azure's management layer, enabling resources to be deployed, organised and governed centrally (Microsoft, 2026b). This supports consistent management while allowing the application to benefit from cloud scalability (Mell and Grance, 2011; Microsoft, 2026b).

In a private-cloud deployment, sensitive components could remain within infrastructure dedicated to the organisation. ARM is not a generic private-cloud manager (Microsoft, 2026b); however, supported on-premises or non-Azure resources

can be projected into ARM through Azure Arc and governed from Azure while remaining outside it (Microsoft, 2025). This can provide greater organisational control, although operational responsibility depends on how the private cloud is managed (Mell and Grance, 2011).

Where sensitive or legacy systems must remain on-premises, hybrid deployment is the most suitable approach for the proposed scenario. The secure web application can run in Azure and be managed through ARM, while sensitive or legacy systems remain in an on-premises private-cloud environment and integrate securely with the Azure-hosted application. This provides flexibility by combining cloud scalability with organisational control over dedicated systems (Mell and Grance, 2011). However, hybrid environments require integration between distinct infrastructures (Mell and Grance, 2011) and introduce additional management complexity (Microsoft, 2025; Muniyandi, Nair and Natva, 2025). Dependence on provider-specific technologies may create vendor lock-in (Opara-Martins, Sahandi and Tian, 2016), while hybrid workload placement can affect performance, resource utilisation and cost in distributed data-processing workloads (Ullah et al., 2024).

Infrastructure Design

The proposed infrastructure uses Azure App Service for compute/application hosting, Azure SQL Database for data storage, and Azure Front Door, VNet and VPN Gateway for networking and hybrid connectivity, as shown in Figure 1.

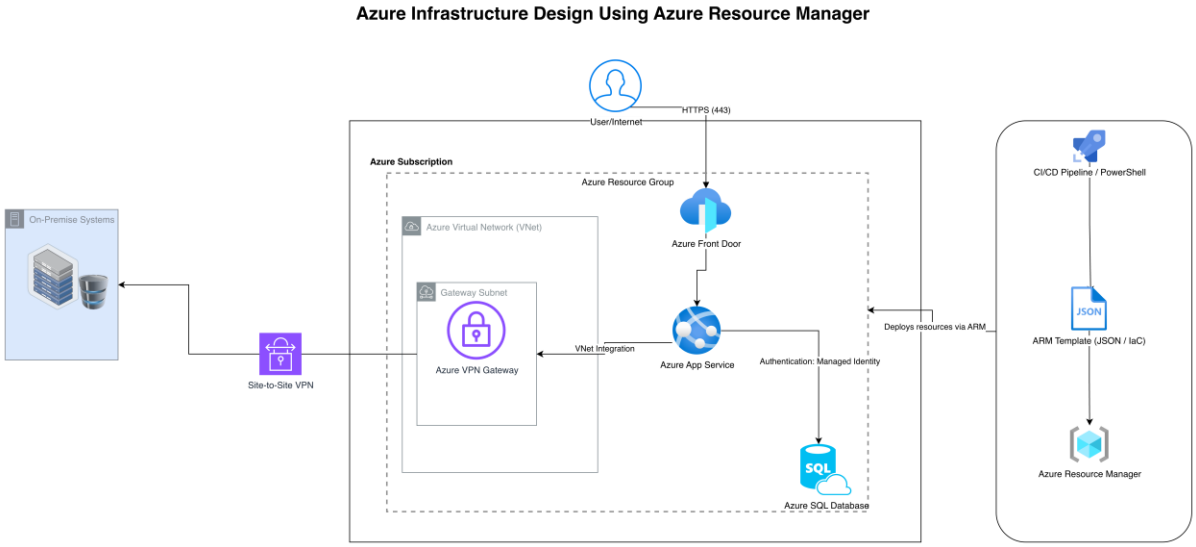


Figure 1: Proposed Azure Infrastructure Design Using Azure Resource Manager

First, the application is configured to use managed identity with Microsoft Entra authentication to connect to Azure SQL Database without storing database credentials in the application (Microsoft, 2026a). Since Azure SQL Database is a managed service, Azure handles the underlying database infrastructure, which reduces the amount of maintenance required from the organisation.

Next, hybrid connectivity is configured so that legacy or sensitive systems can remain on-premises and connect to Azure through a Site-to-Site VPN. The connection terminates at an Azure VPN Gateway inside the virtual network. Azure App Service

then uses VNet Integration to communicate with systems that are still hosted on-premises (Savill, 2019, pp. 181–182, 194–196).

Then, the infrastructure resources are defined in an ARM template written in JSON, including Azure App Service, Azure SQL Database, Front Door and the networking components. Parameters can also be used for values such as deployment region and environment name, allowing the same template to be reused across development, testing and production environments.

Finally, the ARM template is stored in version control and deployed through PowerShell or a CI/CD pipeline. Azure Resource Manager processes the template and deploys the required resources into the resource group. This helps make deployments more consistent and reduces manual configuration between environments (Wurster et al., 2020; Ullah et al., 2021).

Real-World Case Study: Jotun

Jotun provides a real-world example closely aligned with the proposed design. It modernised nine business applications using Azure App Service and Azure SQL Database, supported by Azure security services and integration with on-premises applications (Microsoft, 2021). ARM Templates were used to switch QA and test environments off between releases and restore them when required, helping optimise resources and costs (Microsoft, 2021). Releases increased from two or three per year to every two or three weeks, while software-upgrade downtime fell from hours to 5–10 minutes (Microsoft, 2021). Overall, ARM Templates supported resource and cost optimisation, while DevSecOps practices and App Service deployment slots improved release frequency and downtime; Jotun also reported lower application operating costs (Microsoft, 2021).

Conclusion

Azure Resource Manager provides a consistent way to deploy and manage Azure resources using Infrastructure as Code. Its role differs across IaaS, PaaS and SaaS, while Azure Arc can extend management to some hybrid environments. The proposed design shows how ARM can be used to deploy Azure services together with hybrid connectivity. The Jotun case study also shows how Azure services and automated deployment can improve deployment efficiency and resource use.

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